

Exam Date & Time: 18-Nov-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
THIRD SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2023

Object Oriented Programming [ECM_2124]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Differentiate between thread and process with proper diagram. (2)
 - A)
 - B) Write a Java program to display the HCF (GCD) of two given numbers. (3)
 - C) Create a class Triangle with sides as private fields. Create a subclass RightTriangle with an additional field for height. Similarly, create a class Square and a subclass Cube with appropriate data members. Define an interface with area() and volume() methods, and implement them in all classes. In main(), print area and volume of each shape. (5)
- 2) What is inheritance? Explain types of inheritance supported in Java. (2)
 - A)
 - B) Explain the use of this keyword in Java with an example. (3)
 - C) Write a Java program to: i) Accept a sentence from the user. ii) Create two threads-one to count the number of words, another to find the longest word. iii) After both threads finish, display "TASK COMPLETED". (5)
- 3) Distinguish between compile-time and runtime polymorphism. (2)
 - A)
 - B) Explain how interfaces help achieve loose coupling in Java. (3)
 - C) Create a base class Shape with a display() method. Derive classes Circle and Rectangle. When display() is called using different objects, it should print respective area details. Do not use constructors. (5)
- 4) Write two differences between start() and run() methods of a Thread. (2)
 - A)
 - B) Write a Java program demonstrating use of multiple catch blocks for handling different exceptions. (3)
 - C) Write a Java program where the main thread creates three child threads to calculate the sum of each row in a 3×3 matrix. The main thread displays the total sum of the matrix using results from child threads and also prints its priority. (Use Runnable interface.) (5)

- 5) Explain with an example how a generic method can operate on different types using type parameters. (2)
- A)
- B) Demonstrate the use of an upper-bounded wildcard (< ? extends T>) with a Java Collection. (3)
- C) Create a generic class Pair< T, U> with methods to set and get two values of different types. Write a test program showing pairs of (Integer, String) and (String, Double). (5)

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Event Name	Subject
058-END SEMSTER-ODD-2025	ECM_2124:Object Oriented Programming

#	Course Outcome	Description	Marks
1	1	CO 1	20
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
Total			50

* NA - CO Not Selected

Event Name	Subject
058-END SEMSTER-ODD-2025	ECM_2124:Object Oriented Programming

#	Level	No of Questions	Marks (%)
1	Remember	3	7(14%)
2	Understand	3	8(16%)
3	Application	5	18(36%)
4	Analysis	2	7(14%)
5	Create	0	0(0%)
6	Evaluate	2	10(20%)
Total		15	50 (100%)